

Aabhas Gulati

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RESEARCH INTERESTS

Doctoral candidate at Institut de Mathématiques de Toulouse, Université de Toulouse working in Quantum Information Theory. Interested in mathematical problems in quantum info related to multipartite entanglement, tensors, graphs, optimisation, and symmetry.

EDUCATION

Institut de Mathématiques de Toulouse, Toulouse, France

2023-2026 (Expected)

Doctoral Candidate in Mathematics

- [PhD Thesis](#): Characterizing entanglement constrained by symmetries – Studying geometric aspects of entanglement in symmetric tensors.
- [Supervisors](#): Dr. Clément Pellegrini (Associate Professor, IMT Toulouse), Dr. Ion Nechita (Directeur de Recherche, CNRS, LPT Toulouse)
- [Expected PhD defense](#): 14th September 2026

Indian Institute of Science Education and Research, Mohali, India

2018-2023

BS-MS Dual Degree, Physics (with a minor in Mathematics)

- [MS Thesis](#): Emergence of classicality in quantum resource theories – Studying de Finetti type results in the resource theory of coherence.
- [Supervisor](#): Asst. Prof. Manabendra Nath Bera, IISER Mohali

ARTICLES

Preprints

- [Positive maps and extendibility hierarchies from copositive matrices](#) (with Ion Nechita and Sang-Jun Park)

Published

- [Entanglement in the Dicke Subspace](#) (with Ion Nechita and Clément Pellegrini) in Quantum 10, 2150 (2026)
- [Rank-based entanglement detection for bound entangled states](#) in Phys. Rev. A 113, 022415 (2026)
- [Entanglement in cyclic sign invariant quantum states](#) (with Ion Nechita and Satvik Singh) in J. Math. Phys. 66, 122202 (2025)

PRESENTATIONS

Accepted talks

- Entanglement in the Dicke Subspace (*Quantum Physics and Logic*, Amsterdam 2026, Short Plenary)
- Positive maps and extendibility hierarchies from copositive matrices (upcoming *Theory of Quantum Computation, Communication and Cryptography (TQC)*, Sherbrooke, Canada 2026)
- Detecting entanglement in symmetric states, *Quantum Trajectories*, Bengaluru (2025)
- Graph based approach to detecting bound entanglement, *Young Researcher Symposium, ICMP*, Strasbourg (2024)

Seminar talks

- Entanglement in the Dicke Subspace – QMATH group in University of Copenhagen (May 2026)
- Entanglement in the Dicke Subspace – Séminaire QuantAlps, Institut Fourier Grenoble (May 2026)
- Entanglement in the Dicke Subspace – Séminaire LABRI, Bordeaux (May 2026)
- Multipartite entanglement in mixtures of Dicke states – Grup d'Informació Quàntica (GIQ), UAB, Barcelona (September 2025)

- Multipartite entanglement in diagonally symmetric states – Quantum Information and Thermodynamics group, TU Wien (August 2025)

ATTENDED SCHOOLS/CONFERENCES

- QMATH Masterclass on Representation Theory (August 2025) - Poster: Multipartite entanglement in Bosonic states
- Benasque Quantum Information (July 2025)
- Mathematics and Physics of Quantum Computing and Learning (May 2025) - Poster: Detecting PPT entanglement using rank-one-generated property of matrices
- Random Tensors - IHP, Paris (October 2024) - Poster: PPT Entanglement in symmetric quantum states
- ESQuisses Summer School (June 2024)
- CEQIP (May 2024)
- Workshop QUDATA Bordeaux (January 2024)

ORGANISATION

- Organised the Quantum Cryptography Workgroup in Toulouse (Winter 2025) with Ion Nechita
- Organised the Lean 4 Workgroup in Toulouse (Summer 2025) with Ion Nechita
- Convener of the Physics Club of IISER Mohali (2019-2020)

AWARDS/FELLOWSHIPS

- EUR-MINT PhD Grant, IMT, Toulouse (2023-2026)
- Kishore Vaigyanik Protsahan Yojana (KVPY) fellowship, DST, India – a prestigious national fellowship program funded by the Department of Science and Technology, Govt of India (2018-2023)

SERVICE

I have reviewed articles for Quantum Information Processing (QIP) 2026, Journal of Physics A (co-reviewed with Ion Nechita).

SKILLS & LANGUAGES

- **Programming**: Python, Julia, Mathematica
- **Languages**: Hindi (Native), English (TOEFL 105/120)

RESEARCH STAYS

- QMATH group in University of Copenhagen – Hosted by Alexander Blomenhofer for a week to discuss my work around entanglement and completely positive/copositive tensors (May 2026).
- Institut Fourier, Grenoble – Hosted by Cécilia Lancien for 2 days to discuss mathematical aspects of entanglement in bosonic quantum states (May 2026).

REFERENCES

Dr. Clement Pellegrini | Associate Professor
Probability team,
Institut de Mathématiques de Toulouse
Université de Toulouse III – Paul Sabatier
clement.pellegrini@math.univ-toulouse.fr

Dr. Ion Nechita | CNRS, Directeur de Recherche
Quantum Coherence Team,
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